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Crowdshipping for last-mile delivery in the sharing economy and crowdsourced quick commerce

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Abstract

Purpose – The primary purpose of this research is to fill the gap in understanding crowdshipping by conducting a bibliometric analysis of crowdshipping, mapping its evolution and thematic focus within business research since the concept's emergence. In doing so, we aim to guide the next generation of research and practice.

Design/methodology/approach – In this paper, we performed a structured Scopus search which returned 300 records across 16 subject areas. After filtering documents that we classified as belonging to the Business, Management, and Accounting fields, we used publication trend analysis, citation, and keyword co-occurrence mapping. We then reviewed all documents to understand the contributions they made to the field and to identify the gaps that future research should address.

Findings – Our literature analysis showed that scholarly output began in 2017, surged during the COVID-19 pandemic, and remained concentrated in North America and Western Europe. It reveals the main research streams, including business models, operational optimization, market behavior, sustainability assessments, and technology support. It also identifies issues remaining to be addressed, such as inconsistent terminology use, regional research gaps, and methodological limitations. Among the practical outcomes of this work is its literature-driven last-mile delivery (LMD) strategy canvas.

Originality/value – This is one of the first literature reviews focused exclusively on crowdshipping in business research, offering a comprehensive and structured direction for future research in the business field. It also offered an LMD strategy canvas for industry.

Keywords Crowdshipping, Crowd shipping, Crowdsourced delivery, Collaborative logistics, Sharing economy logistics, Last-mile delivery strategy canvas

Paper type Literature review

1. Introduction

Electronic commerce (E-commerce), which is the online buying and selling of goods or services via the internet, has grown significantly. Customer expectations have also increased, and include faster and cheaper deliveries (Jain *et al*., 2021; Sulubara *et al*., 2025). In the E-commerce sector, the COVID-19 pandemic and the subsequent control measures like social distancing and lockdowns resulted in increased competition for shorter delivery time windows, leading to the development of innovative new delivery systems (Miljenović and Beriša, 2022; Ranjekar and Roy, 2023). In the product delivery context, particularly with the regard to Last-Mile Delivery (LMD), crowdshipping is a system in which ordinary people make final deliveries from stores or warehouses to customers' homes (Bajec and Tuljak-Suban, 2022; Mittal *et al*., 2022; Upadhyay *et al*., 2021). Alnaggar *et al*., (2021) find that several start-ups have emerged in recent years that offer crowdshipping platforms, the goal of which is to make shipping more affordable than traditional postal services or high-priced

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couriers. These crowdshipping platforms are part of the broader emerging concept of the sharing economy, which has produced successful business models over the last decade. Crowdshipping connects shippers to a network of people who are willing to provide shared-mobility services as a side business to supplement their income.

The initial goal of LMD is to reduce retailer delivery costs, but it may also have environmental benefits if performed using zero-emission transportation or when taking advantage of existing trips with minimal detour. Thus, crowdshipping benefits society when done correctly by reducing the number of freight delivery trucks in urban/suburban areas, allowing businesses to cut delivery costs while maintaining the same level of service. This novel design also promotes social collaboration, allowing ordinary people to participate in LMD practices and contribute to a more sustainable community.

However, crowdshipping is difficult due to its novelty, lack of practice, and complexity. To put in place an effective and efficient crowdshipping system, it is necessary to understand what academic and scholarly investigations and examinations have found. Scholarly contributions have the ability to clarify new business operations values, as Sorooshian (2024) noted. Business research performance, as stated by (Snihur and Bocken, 2022), affects business ecosystems, societies, and the entire planet. The application of its lessons can lead to new developments (Sorooshian, 2024). As a result, the aim of this study is to conduct a quantitative analysis of the academic literature on crowdshipping, and as such it is one of the earliest literature reviews on this topic.

Since most studies in this area have been limited to concentrating on specific concerns, gaps exist in the literature. First, the lack of standardization among the terms used to describe this field contributes to the confusion that surrounds it. The absence of agreement on terminology (e.g. Mak, 2025; Mohri *et al.*, 2025a; Punel and Stathopoulos, 2017) prevents researchers from integrating their results. Second, due to behavioral differences in different geographic locations, the concentration of crowdshipping research in a specific part of the world presents challenges for generalizing findings and conclusions (Le and Ukkusuri, 2019). Third, although most studies on crowdshipping have been recent, many unanswered questions remain unaddressed by researchers (Alnaggar *et al.*, 2021; Pourrahmani and Jaller, 2021). Despite recent papers highlighting the importance of field-based research that incorporates behavioral aspects into optimization, the existing body of empirical evidence lacks long-term real-world data. Although research on crowdshipping business operations has been requested (Le and Ukkusuri, 2019; Pourrahmani and Jaller, 2021), very few researchers have attempted to provide it. As a result, the key objective of this study is to provide a roadmap for scholars to address these structural limitations and the lack of integrative research. Consequently, the following three questions are those that this paper seeks to answer:

(1) How is the business research literature on crowdshipping structured?

(2) What are the scholarly contributions to the field?

(3) What are the most appropriate avenues for future research?

To answer these questions, the rest of the article is organized as follows. The next section covers the crowdshipping revolution and provides necessary background on the sharing economy and LMD. Section 3 explains the methods we used for answering the research questions identified above. Section 4 presents our findings and Section 5 discusses them, outlining their implications for research, practice, and society. The paper ends with Section 6, which offers our conclusions.

2. Background

2.1 LMD in e-commerce

Recent decades have seen a focus on gaining competitive advantage in supply chains, driven by advances in information technologies and the internet's impact on the business

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environment, necessitating dynamic network cooperation in supply chain management (SCM) (Nikakhtar and Jianzheng, 2012). Current SCM, in which logistics accounts for a considerable portion of the overall cost (Lukinskiy *et al*., 2015), has significantly evolved since the emergence of e-commerce (Alsheyadi *et al*., 2024). For example, revenue from online food delivery has increased by 7.5% in the UK, 7.7% in Germany, 8.4% in Italy, 10.6% in France, and 10.7% in Spain (Kumar and Chidambara, 2024).

The COVID-19 pandemic accelerated the E-commerce logistics transition (Kumar and Chidambara, 2024; Olawade *et al*., 2024; Villa and Monzón, 2021). E-commerce logistics has two major business models: business-to-consumer and business-to-business (Mangiaracina *et al*., 2015; Uvet *et al*., 2024). For both, logistics typically consists of three major phases: replenishment of goods from manufacturers to distribution centers; order fulfillment at distribution centers via sortation, packing, and picking operations; and LMD to buyer(s). The term "last-mile" originated in the telecommunications industry and is now used to describe the final stage of a delivery process (Lim *et al*., 2018).

Nonetheless, modern logistics methods are crucial for improving material flow efficiency and reducing distribution costs, while e-commerce growth expands the logistics market and drives the development of logistics technology (Gomes *et al*., 2023; Lim *et al*., 2018). The rise of e-commerce and the new normal is rapidly increasing the demand for better LMD (Nandy and Padhy, 2025; Sharma *et al*., 2025). It has significantly increased the operations' complexity, as customers demand faster and more flexible delivery methods (Yu *et al*., 2017). The LMD of logistics phases has become a critical source of market differentiation, prompting retailers to invest in a variety of consumer delivery innovations, including (but not limited to) parcel lockers, autonomous delivery solutions, crowdshipping, and other alternative handover options (Mangiaracina *et al*., 2019). Thus, LMD has become a critical component for businesses, despite being previously considered a supporting capability (Yu *et al*., 2017).

Yu *et al*., (2017) explain that the unprecedented demand for home deliveries is expected to result in traffic congestion and increased carbon emissions. In many cases, the LMD is managed by a dedicated third party (Sharma *et al*., 2025), but today, LMD is perceived as an inefficient, environmentally harmful, and expensive element of the supply chain in terms of transportation use (Zosu *et al*., 2024). As a result, scholars (Lim *et al*., 2018; Oliveira *et al*., 2017) emphasize the importance of a comprehensive understanding and improvement of LMD.

2.2 The sharing economy and its relevance to today's LMD

The logistics sector is now modernizing to meet customer demands, driven by the growing acceptance of e-commerce (Elia *et al*., 2024). The sharing economy paradigm moves away from ownership-based models and toward use-based models, facilitating better resource utilization and access to a pool of users (Elia *et al*., 2024; Govindan *et al*., 2020). This paradigm connects people via platforms to conduct sales, rentals, swaps, or donations; create and share goods, services, space, and money; produce, access, and circulate resources; and centralize access to idle assets (Görög, 2018; Moncef and Monnet Dupuy, 2021). Crowdsourcing, which is often managed via smartphone apps and online platforms, stands out among sharing economy models because it allows individuals to contribute their logistical resources and skills (Alromema *et al*., 2025). Thus, the sharing economy is now transforming logistics sectors and business models, with a large portion of it generating the physical flow of goods, increasing deliveries, and broadening the scope of the logistics resources being mobilized (Moncef and Monnet Dupuy, 2021). The economy is shifting to a sharing economy model that optimizes resource utilization and facilitates transactions through intermediaries, as a result of e-commerce and personalized demands in the logistics sector (Upadhyay *et al*., 2021). Recent research has begun to identify a promising area of study at the intersection of the sharing economy and logistics (Carissimi and Creazza, 2022; Matusiewicz and Książkiewicz, 2023).

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The sharing economy's participation in logistics occurs at every stage of the logistics value chain. Risk liability, insurance, transparency, and workforce protection are the building blocks of its models (Saglietto, 2021). In warehousing, as explained by Saglietto (2021), the sharing economy enables shared utilization of and billing for existing customer warehouses. Sharing economy platforms have also investigated crowdshipping initiatives for their ability to generate value and provide faster and more flexible LMD services in logistics, enable cost optimization, and reduce transportation carbon footprint (Upadhyay *et al*., 2021). Despite the sharing economy's potential, its economic, social, and environmental consequences remain largely unknown (Moncef and Monnet Dupuy, 2021). More broadly, Saglietto (2021) argues that more research support for digitalizing supply chains based on a sharing economy collaborative system is critical for achieving Logistics 4.0, which incorporates advanced information and communication solutions to achieve interoperability, real-time communication, and efficient LMD.

2.3 The crowdshipping revolution

In practice crowdsourcing is not new (Mladenow *et al*., 2015a), but Mladenow *et al*. (Mladenow *et al*., 2015b) claimed that this concept had begun to excite both academic and practical interest close to 2015. The rapid growth of e-commerce has had a significant impact on the retail and logistics sectors (Noor *et al*., 2023). Consumers now seek personalized service, flexible delivery options, and convenient parcel collection and return methods; these are driving this effect (Vakulenko *et al*., 2019). The logistics sector is facing increased resource demands: to address the challenges, LMD actors are looking for new delivery service solutions (Mangiaracina *et al*., 2019). Many consumers prioritize delivery times and shipping costs as the most significant factors influencing their online purchase decisions (Harter *et al*., 2024). The quick commerce (Q-commerce) business model promises extremely rapid product delivery, allowing users to expect their orders to arrive on their doorsteps in minutes, satisfying urgent demands. The rapid expansion and increasing accessibility of Industry 4.0 ideas, alongside the digital technologies, increasing demand, and anti-epidemic measures enacted during the COVID-19 pandemic, have collectively fostered the emergence of Q-commerce as an innovative development in e-commerce (Stojanov, 2022). Nonetheless, the motivation for innovation is to contribute to address challenges in society (Boon and Edler, 2018).

Q-commerce is experiencing explosive growth (Ranjekar and Roy, 2023).With the growth of the Q-commerce industry, the LMD sector is expected to experience another crowdsourcing boom. "Crowd shipping" or "crowdshipping" can cut the costs and delays associated with LMD (Dayarian and Savelsbergh, 2020). It is the use of social networking to encourage collaborative behavior, such as sharing services and assets for the benefit of the community and oneself (Buldeo Rai *et al*., 2018). It uses surplus capacity on planned trips to speed up deliveries, improving LMD efficiency while reducing emissions and traffic congestion (Buldeo Rai *et al*., 2017; Simoni *et al*., 2020); it emphasizes the use of unused carrying capacity in any ordinary individual's transportation means to carry goods for others (Buldeo Rai *et al*., 2017) via the sharing economy concept. It is gaining popularity due to the ease of hiring crowdshippers (Macrina *et al*., 2024), and it is a reliable choice for tackling the rising demand for e-commerce LMD (Rossolov and Susilo, 2024). The necessarily brief history of the practice may leave business adopters skeptical about the success of these solutions and the risks they entail (Catalini and Tucker, 2017); however, recent research has shown that individuals are even willing to pay fees for crowd-shipped delivery of their online purchases (Rossolov and Susilo, 2024), indicating market demand and readiness for such service.

Vakulenko *et al*., (2019) argue that a lack of knowledge can result in new service implementation plans being based on trial-and-error methods, lowering investment returns and potentially causing customer loss due to changes in traditional delivery services. Sorooshian (2024), however, argues that business research is responsible for expanding understanding and supporting new adapters until general awareness emerges from practice.

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The growing gig economy is another driving force behind the crowdsourcing revolution. Gig workers are people who take temporary jobs completing tasks or providing services for a set period of time (Myhill *et al*., 2023). Gig workers' share of the workforce is growing quickly; Ray *et al*., (2024) projects that it will reach 40% in 2025. Society therefore possesses sufficient crowd shippers to fulfil consumer demand for crowdshipping LMD. Although crowdshipping appears to be among the main solutions for building modern societies and the societies of the future (Capponi *et al*., 2019), its terminology is not yet standardized and academics may describe it using various terms (Mohri *et al*., 2023; Saglietto, 2021), including crowd shipping, crowdshipping, crowd-sourced shipping, crowd-sourced delivery, sharing economy logistics, crowd logistics, collaborative logistics, and cargo hitching. Researchers continue to contribute to its study, although the current development of this field is not fully integrated and continues to leave certain concerns unresolved (Çelik *et al*., 2025; Le and Ukkusuri, 2019; Pourrahmani and Jaller, 2021). As a result, the disorganized nature of the existing literature motivates us to undertake this study to map the structure of past crowdshipping studies, synthesize their contributions, and direct the next generation of business research.

3. Methodology

This study's research questions are addressed through bibliometric analysis, which is a quantitative approach to analyzing research trends, networks, and patterns. Because it allows to map the evolution of crowdshipping and LMD research in a transparent and replicable manner, a bibliometric review is chosen in this article. Besides, the paper supplements the quantitative mapping with a qualitative synthesis of the included studies to interpret the main contribution streams and research gaps from a business and management perspective.

Two of the primary scientific databases used for bibliometric analysis are Web of Science and SCOPUS; however, Scopus is generally preferred as it offers more extensive coverage (Sorooshian, 2024; Sorooshian *et al*., 2022; Sorooshian *et al*., 2023a). Hence, we conducted a search for existing literature in the Scopus database (www.scopus.com) on 8 May 2025, covering all years available up to that date. To capture the fragmented and heterogeneous terminology that characterizes this research area, the search string includes a broad set of competing terms and spellings. However, to avoid language barriers, only English documents were considered for analysis. Hence, to search Scopus, we used the search inquiry of "TITLE-ABS-KEY ("Crowdshipping" OR "Crowd shipping" OR "crowd-shipping" OR "crowd-sourced delivery" OR "crowd sourced delivery" OR "sharing economy logistics" OR "crowd logistic*" OR "crowd-logistic*" OR "crowd-sourced shipping" OR "cargo hitching") AND (LIMIT-TO (LANGUAGE, "English"))." Following the recommendation made by Tavana and Sorooshian (Tavana and Sorooshian, 2023), we used the asterisk symbol (*) as a placeholder to represent one or more characters in a search query.

The search yielded 300 documents across 16 subject areas. The five subject areas that contributed the largest number of documents were social sciences (with 150 documents); engineering (with 134 documents); computer science (109 documents); business, management, and accounting (75 documents); and decision sciences (70 documents). These were followed by mathematics (54 documents); energy (31 documents); environmental science (27 documents); economics, econometrics, and finance (14 documents); and smaller subject area contributors (with just 1 or 2 documents).

Our study explores crowdshipping as a managerial innovation in the e-commerce sharing economy. Excluding technical (computer science/engineering or mathematical sciences) or socio-environmental-political (social sciences, economics, energy, or environmental management) documents, our analysis focuses on crowdshipping from a business-level and managerial standpoint. To achieve this focus, and similar to Sorooshian *et al*. (Sorooshian *et al*., 2023b) in their study of business research, we directly filtered the Scopus search results into the subject area "Business, Management, and Accounting", as it is most relevant to this

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paper's scope. We thus limit the bibliometric analysis to the 75 documents classified by the Scopus database as business research indexed contributions.

The overall process follows the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) main stages (identification, screening, eligibility, inclusion (Selçuk, 2019)) and is summarized in Figure 1.

We then use this business research-restricted dataset in all subsequent publication trend observation, citation and keyword occurrence, and abstract analyses. To achieve this, we used Scopus analytics tools, ChatGPT (https://chatgpt.com/) artificial intelligence version o4-mini-high, and ScienceSpace (https://medialab.github.io/sciencescape/) to interpret the data. Finally, we reviewed the title and the abstract of all the documents in order to summarize the main contributions they made.

4. Results

This section is organized into four subsections. The first three subsections address the research questions. The last subsection, however, proposes an LMD strategy canvas.

4.1 Business research structure with regard to crowdshipping

Scopus classified 67 of these 75 documents as research articles, four as conference papers, two as book chapters, and two as literature reviews. The documents listed more than 150 unique keywords, and as expected, the most-used keywords aligned well with the scope of this research. The distribution of the available Scopus-indexed crowdshipping-related business research is presented in Figure 2. As the figure illustrates, the first relevant Scopus-indexed documents are from 2017. As we collected the Scopus data in early 2025, the previous year's data were not yet completely compiled (meaning that the publication record will grow). They consist of six documents: one conference paper and five research articles. Researchers from the United States and Western Europe are the founders of the concept in business research, producing all of the indexed works in the first two years.

Approximately a decade of business research effort, from 2017 to the present, reveals shifts in academic interests. After 2017, the interest level fell slightly until 2019: there are only three

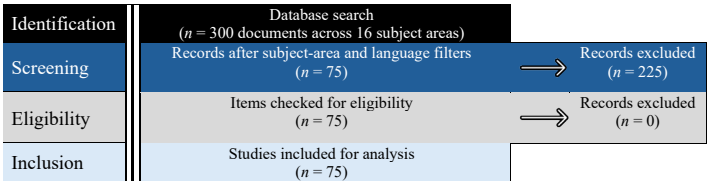


Figure 1. Data identification, screening, eligibility, inclusion

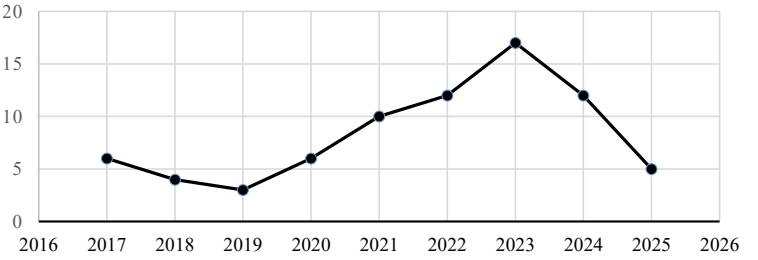


Figure 2. Annual publications

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indexed works for 2019. However, as the COVID-19 pandemic began, more publications were produced and indexed annually, with a record of 17 documents appearing in 2023. A decline is visible in 2024, which could be attributed to Scopus indexing lag or may be the result of other reasons.

According to the records, multiple publishers participate in the dissemination of scholarly knowledge in this domain. However, Elsevier hosts three of the top four journals, while Emerald Publishing hosts the other. Moreover, the geographical distribution of the authors reveals 35 countries contributing authors to these identified documents. Beyond the countries represented there are notable regional gaps: Central and South America, the majority of African nations, the Middle and Near East, the ASEAN bloc, and Russia are all absent from the list. The United States leads the field with 16 indexed documents and several research contributors; Italy (with 9 documents), France (8 documents), and Germany (8 documents) come next.

4.2 Business research contributions to the crowdshipping discussion

Figure 3 depicts how the research that has been conducted feeds into subsequent work. The citation trend indicates that the collected documents aided other researchers and facilitated the advancement of crowdshipping discussions. The funnel shape demonstrated by the figure also shows how the collection of data benefits not only business research but also other subject areas.

Figure 4 illustrates the developing usage of the most significant keywords (i.e. those that appear in more than 10 papers). The numbers in parentheses after the keywords indicate the number of papers utilizing those keywords. This figure indicates that the terminology employed in the discussions lacks standardization; however, “crowdshipping” and “crowd logistics” are predominant. The graphic shows that the term “crowd-shipping” appeared in some recent documents, as evidenced by the darker sections on the right side. However, the term “crowdshipping” has been used since earlier.

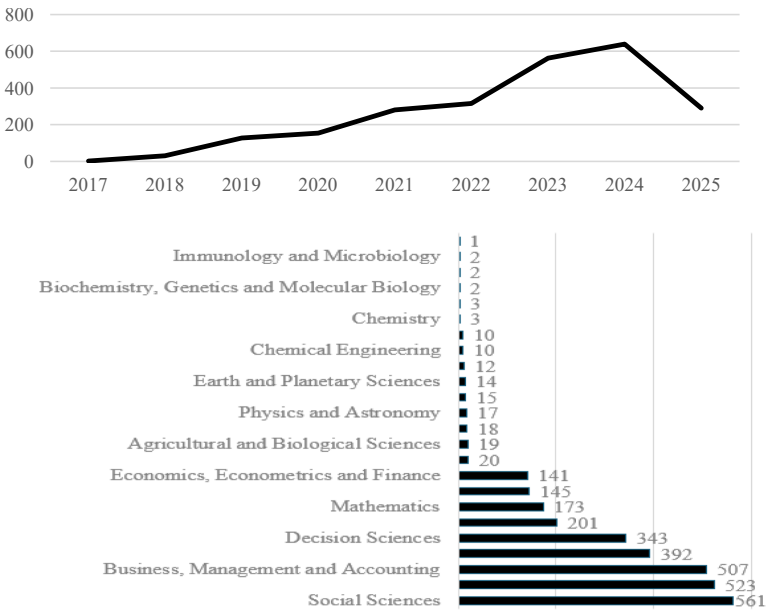


Figure 3. Citation growth

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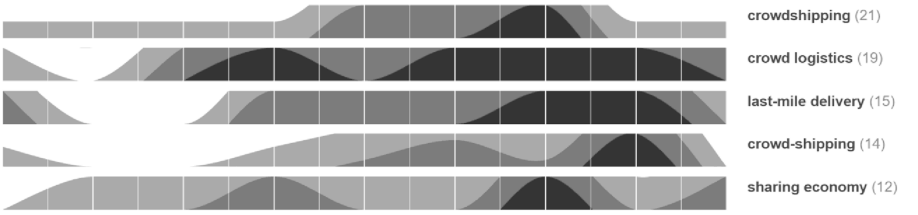


Figure 4. Keywords evolution

Between 2017 and 2025, the research landscape for "crowd-" and LMD expanded and evolved in a variety of directions. The "top keywords per year" analysis reveals that the field has evolved from basic routing and locker-based solutions to sharing-economy models. It now includes detailed analyses of consumer behavior, agent-based systems, and experimental choice methods, while retaining a focus on core optimization and scaling up crowdshipping paradigms.

From 2017 to 2019, the emphasis was on conducting fundamental research while investigating method-based approaches. The initial 2017 research focused on defining major delivery challenges using "crowd-" terminology (e.g. crowdshipping, crowdsourcing, crowd logistics, etc.) and traditional mathematical programming methods. In 2018, the industry introduced terms like "sharing economy," "value co-creation," and "alternative delivery points" to describe new business models and entry points. The introduction of simulated annealing and content analysis methods in 2019 signaled a consumer-focused logistics transformation that included analytical frameworks, collaborative consumption, and CDPs (Collection and Delivery Points). Between 2020 and 2022, the field grew through methodological diversification.

The research fields of "crowd logistics" and "crowdshipping" gained significant traction during this period, with the number of crowd logistics papers increasing threefold in 2020 and fourfold in 2021. Agent-based modeling/simulation entered the field in 2020, and routing regained importance in 2022. The field's focus shifted from algorithmic studies to systems-level and human-behavioral research as "crowdsourced delivery" and urban-oriented terms (city/urban logistics, urban freight) gained traction.

The period from 2023 to 2025 saw maturity and increased nuance enter the field, which expanded to explore new frontiers. The number of crowdshipping papers peaked at six in 2023, while detailed behavioral methods such as stated choice/preference experiments appeared, indicating an increase in user acceptance research. The term "last-mile logistics" emerged as a separate category from "last-mile delivery," demonstrating the development of more precise categorization. The number of "crowd-" papers fell in 2024, but vehicle routing remained strong alongside urban freight and logistics research. Early 2025 saw rapid diversification in the field as researchers investigated crowd logistics and the sharing economy, as well as new topics such as supply chains and action space design, indicating advanced system integration and decision frameworks.

It is evident that business research on crowdshipping has produced a substantial body of work that includes various conceptual, methodological, and empirical contributions. Reviewing the abstracts of the 75 identified documents provides several interconnected streams, as listed in Table 1, showing how business researchers have contributed to the topic.

1- Business Models: The initial research established fundamental concepts that resulted in the classification of crowdshipping business models. For example, one study transformed expert interview data into four implementation steps, while also generating open research questions about sustainable crowd logistics services (Frehe *et al.*, 2017). A subsequent taxonomic study (Ciobotaru and Chankov, 2021) examined more than 100 crowdshipping companies using cluster and principal component analyses to identify six distinct archetypes

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Table 1. Business research streams

Contribution stream	Sample representative documents
1- Business models	Cieplińska and Szmelter-Jarosz (2020), Ciobotaru and Chankov (2021), Dablanc *et al*. (2017), Frehe *et al*. (2017), Le *et al*. (2021), Michel *et al*. (2023), Roussat *et al*. (2023), Szmelter-Jarosz *et al*. (2023), Zenezini *et al*. (2022)
2- Operational optimization	Allahviranloo and Baghestani (2019), Dayarian and Savelsbergh (2020), Farazi *et al*. (2022), Huang *et al*. (2022), Hwang *et al*. (2025), Mohri *et al*. (2025b), Pugliese *et al*. (2023a), Pugliese *et al*. (2023b), Tadić *et al*. (2023), Triantali *et al*. (2025), Triki (2021), Wyrowski *et al*. (2024), Yang *et al*., (2024), Yıldız (2021), Yu *et al*. (2022)
3- Market behavior	Cebeci *et al*. (2023), Ermagun and Stathopoulos (2021), Fessler *et al*. (2022), Giglio and Maio (2022), Mohri *et al*. (2024), Nguyen *et al*. (2023a, b), Punel and Stathopoulos (2017), Rossolov and Susilo (2024), Shen and Lin (2020), Wang *et al*. (2023), Wicaksono *et al*. (2022)
4- Sustainability concerns	Akeb *et al*. (2018), Awasthi *et al*. (2025), Chen and Chankov (2017), Fessler *et al*. (2022), Melkonyan *et al*. (2020), Mittal *et al*. (2021), Nascimento *et al*. (2023), Rai *et al*. (2018), Wyrowski *et al*. (2024)
5- Technology supports	Alqaisi *et al*. (2023), Castillo *et al*. (2022), Chua *et al*. (2025), Cramer and Fikar (2024), Le *et al*. (2021), Sawik (2024), Zenezini *et al*. (2022)

and determine which configurations customers find most appealing. These frameworks provide tools for scholars and practitioners to evaluate, analyze, and develop crowdshipping strategies.

2- Operational Optimization: A significant part of the business research has focused on modeling and solving various variants of the vehicle routing problem that apply to crowdshipping contexts. One study, for example, shows that adaptive variable neighborhood search procedures designed specifically for grocery delivery can efficiently solve cases involving up to 200 orders (Hwang *et al*., 2025). The combination of set-partitioning formulations and decomposition heuristics enables the solution of city-scale crowdshipping instances with 1,000 orders in minutes (Yang *et al*., 2024). Deep reinforcement learning agents that use constrained double-dueling architectures with heuristics integration have 24–37% lower shipping costs than traditional heuristics in dynamic on-demand scenarios (Farazi *et al*., 2022). Collectively, algorithmic innovations have pushed the limits of real-time large-scale crowdshipping optimization.

3- Market Behavior: Business research also focused on analyzing willingness-to-pay together with participation intentions and perceived risks as a growing field of empirical investigation. Bicycle crowdshipping market analyses, for one, show how demand-supply functions meet in real-world settings (Wicaksono *et al*., 2022). The effect of trust on the choice for crowdshipping services (Cebeci *et al*., 2023) also demonstrates that trust acts as a complete mediator between adoption and reputation and damage concerns while acting as a partial mediator for time and price attributes through stated-choice experiments that include trust as a latent variable. These business research insights map user segments and inform tailored incentive and communication schemes.

4- Sustainability Concerns: Crowdshipping's combined effects on environmental systems and social structures have also been investigated in business research. One case study shows that crowdshipping without proper matching systems and incentives incurs higher external transport costs than traditional parcel services (Rai *et al*., 2018). Current supply chains' integration with crowdshipping and digital coordination tools using system dynamics frameworks demonstrates how it reduces carbon emissions while increasing delivery reach (Chen and Chankov, 2017). According to qualitative research (Akeb *et al*., 2018), crowdshipping requires user-centered design and social value rather than monetary rewards to sustain volunteer-based LMD. Long-term benefits depend on advanced platform development, as well as proper regulatory control and stakeholder training.

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5- Technology Supports: Business research on platform and technology features has revealed how tipping mechanisms, reputation systems, and incentive structures influence agent behavior. The combination of netnography-based adaptive multi-agent simulations, for example, demonstrates that customer tips reduce delivery uncertainty and operational costs, but the effects depend on urban population density. The issue of increased agent independence poses new challenges for crowdsourcing technology platforms specializing in on-demand LMD (Castillo *et al*., 2022). A decentralized LMD framework could use blockchain-enabled architectures to promote governance democratization and lower intermediary costs (Alqaisi *et al.*, 2023). The integration of on-demand mobility fleets with bilevel pricing games produces double-digit cost savings and profit uplifts when fees reach their optimal levels (Le *et al*., 2021).

4.3 Possible directions for future business research

75 indexed documents in nearly ten years do not represent a mature area of knowledge; more work in this field is undoubtedly required. More mixed-methods field research, scalable pilot programs, and integrative evaluation frameworks are required in order to understand how best to promote responsible crowdshipping and close existing knowledge gaps. Additionally, section 4.1 of this paper revealed that inconsistencies in terminology exist that could prevent proper knowledge accumulation. Researchers could follow what we have revealed as the latest trend and use the "crowdshipping" term. More conceptualization and literature review articles could go beyond existing taxonomies to establish core definitions, dimensions, and typologies. Section 4.1 also revealed that scholarly publications are currently concentrated in North America and Western Europe. Researchers, particularly in Latin America, Africa, and Asia, require additional research to account for differences in infrastructure, business and social environments, and market norms. Academics from countries that have as yet contributed less to the discourse should form transnational partnerships with those from more research-producing areas so that they can co-design pilots, share datasets, and create solutions that take local needs into account while advancing global theory.

The literature corpus, as explained in section 4.2, is broad in scope, including both operational algorithms and strategic taxonomies, but it lacks cross-validation. Sustainability assessments often concentrate on isolated locations or cases; many algorithmic studies conduct research using synthetic benchmarks and small-scale simulations, which limits their ability to permit researchers to generalize their findings. Few larger-scale studies exist, and they have limitations. One example tested an adaptive larger search algorithm using generated grocery datasets, but the city-scale decomposition heuristics were validated using only simulated trip requests (Hwang *et al*., 2025). To ensure robustness, future research should include field pilots that validate results across multiple existing settings and demand profiles.

More research is needed to compare developments in real-world settings, including high-density Asian cities, the sprawling society architecture of North American, and mixed-mode European environments. Moreover, the majority of studies considering market behavior are based on surveys taken at a single time point. These taxonomies fail to account for the rapid changes in platforms and regulations. Many studies (such as (Cebeci *et al*., 2023; Wicaksono *et al*., 2022)) have demonstrated trust effects in one context, but it is unclear how preferences change as platforms evolve. Behavioral studies must use more longitudinal field experiments, or multiple deployment phases, to see how preferences change after implementation. The business model research steam has advanced our conceptual framing, but labor protections, business law frameworks, and data privacy are still neglected.

4.4 LMD strategy canvas

The need for better connections between academic research and business practice prompted us to create an LMD strategy canvas, which combines five research streams into six decision dimensions. We present this canvas in Figure 5. It assists managers by using targeted

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Figure 5. LMD strategy canvas

questions, soliciting evidence-based insights and targeted actions. This structure enables managers to diagnose their current capabilities and prioritize investments while running targeted pilots and tracking performance using empirical findings, simulation, and field experiment validation.

It outlines a practical strategy for enhancing LMD in logistics. The strategy encompasses customer segmentation, crowd sourcing archetype selection, optimization implementation, the prioritization of high-impact technologies, sustainability, and the monitoring of performance indicators. As a result, this canvas can expedite decision-making and create a longitudinal evidence foundation for improving LMD strategy and enabling scalable, sustainable, and customer-focused logistics.

4.5 Discussion of the findings

The rapid expansion of e-commerce, the sharing economy, and Q-commerce, as well as the pressures of the COVID-19 era, have propelled crowdshipping into the forefront of LMD logistics. Our findings reveal that crowdshipping research has evolved into multiple subfields while remaining limited to specific geographical areas.

Our study expands on previous studies on sharing economy logistics conducted by Saglietto (2021) and Mohri *et al*., (2023) by focusing on business-oriented crowdshipping research and evaluating intellectual and geographical advancements. Our research combines five distinct research streams into an operational LMD strategy canvas that includes sustainability concerns, business models, market behavior, operational optimization, and technology support. The existing research contributions are based on restricted simulation studies and single-site surveys (Wicaksono *et al*., 2022; Cebeci *et al*., 2023), but our findings show that future studies will need to conduct multi-city field experiments to generate robust findings applicable across different contexts. Our work's cumulative results combine theoretical foundations with practice-based requirements.

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The social consequences of crowdshipping operations require immediate attention because, in addition to their benefits, they pose significant problems. Crowdshipping platforms, when implemented correctly, reduce carbon emissions and urban traffic congestion by utilizing available capacity from scheduled trips (Buldeo Rai *et al*., 2017; Gatta *et al*., 2018; Simoni *et al*., 2020). The platform allows users to participate in LMD operations, generating new revenue streams for gig economy workers (Ray *et al*., 2024). Therefore, platform operators and policymakers must quickly address the social issues arising from crowdshipping operations. These include the unpredictable nature of their work, the lack of standard operating procedures, and privacy and trust concerns (Cebeci *et al*., 2023; Nguyen *et al*., 2023a). Moreover, the operational efficiency of crowdshipping creates new societal, environmental, and governance challenges that must be addressed immediately (Qi *et al*., 2018; Rai *et al*., 2018). The research suggests that, like any other business, crowdshipping businesses must keep their costs low while providing dependable services to their customers (Dehghan *et al*., 2025; Mousavi *et al*., 2022).

The LMD strategy canvas we developed in this research provides decisionmakers with a structured approach to launching pilot programs, assessing technology deployment, and monitoring sustainability metrics. This study demonstrates that policymakers must develop expedited regulatory systems that protect gig workers' rights while promoting environmentally friendly delivery solutions and facilitating global knowledge-sharing in order to bridge the knowledge gap between research and practice in various regions. In short, the study identifies three critical directions for future research: (1) expanding geographical coverage, (2) diversifying methodological approaches, and (3) conducting longitudinal investigations. As a result, we believe that this article will contribute to the development of new theoretical contributions while also providing managers and regulators with practical solutions for creating environmentally friendly and accessible LMD systems.

5. Conclusion

Crowdshipping is the outsourcing of logistics services, via the mediation of matchmaking platforms, to a large number of dispersed, on-demand agents, with each agent being compensated for the LMD (usually with a compensation fee). In this context, this paper sought to answer: (1) How is the business research literature on crowdshipping structured? (2) What are the scholarly contributions to the field? And (3) what are the most appropriate avenues for future research? It also proposed an LMD strategy canvas that deserves to be reviewed by other researchers and practitioners.

In this paper, we argued that future business research should be based on practice needs. This work, serving as a role model, guides future literature reviews on topics related to the role of the sharing economy in logistics: crowd warehousing, for instance, should be the subject of a separate study. Our study used bibliometric data, focusing on published business research documents indexed in the Scopus database. While this provides a curated and retrospective dataset, future literature review papers could benefit from incorporating data from other databases to generate a more comprehensive overview. Future research should also consider incorporating other research subject areas in addition to business research to offer a more extensive understanding of the crowdshipping context.

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